

Yilang Liu

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SUMMARY

Yale Robotics Ph.D. candidate developing synthetic data pipelines, VLA policies, and sim-to-real learning methods. Deployed $\pi_{0.5}$ in **18/20 successful factory trials** and transferred visual policies zero-shot to real robots. Publications: NeurIPS 2025 (**Spotlight**), CoRL 2026, RSS 2026, and ICRA 2026.

EDUCATION

Yale University	New Haven, CT
<i>Ph.D. Candidate in Mechanical Engineering (Robotics & Robot Learning)</i>	<i>Expected May 2027</i>
Carnegie Mellon University	Pittsburgh, PA
<i>M.S. in Mechanical Engineering (Research Program)</i>	<i>May 2022</i>
University of Cincinnati & Chongqing University	Cincinnati, OH / Chongqing, China
<i>B.S. in Mechanical Engineering (cum laude)</i>	<i>Apr 2020</i>
<i>B.E. in Mechanical Design, Manufacture and Automation</i>	

INDUSTRY EXPERIENCE

Dexmate Inc. (<i>bimanual mobile-manipulation robotics startup</i>)	Santa Clara, CA
<i>Robotics Intern</i>	<i>Oct 2025 – Jan 2026</i>

- Fine-tuned and deployed Physical Intelligence’s $\pi_{0.5}$ (pi0.5) VLA for bimanual condenser placement onto a conveyor, achieving **18/20 successful on-site factory trials**; stabilized execution at **~20 Hz** with Real-Time Action Chunking (RTC).
- Post-trained NVIDIA’s GR00T N1.6 VLA on [Dexmate Vega](#) for a long-horizon (**2-minute**) bimanual PVC-box unfolding-and-placement sequence, reaching **8/10 successful on-robot trials** after the zero-shot model failed.
- Collected and curated ~80 teleoperated demonstrations in LeRobot format; built the on-robot evaluation protocol and the two-GPU distributed fine-tuning pipeline (torchrun).
- Trained and deployed an ACT policy for vision-based bimanual bottle grasping and placement; developed an RRT-guided PPO policy for 6-DoF dexterous grasping in Genesis and Isaac Sim.

SELECTED RESEARCH PROJECTS

Graduate Research Assistant, Yale University	New Haven, CT
<i>CoCuRo Lab; Advisor: Prof. Ian Abraham</i>	<i>Aug 2022 – Present</i>

EmbodiedSWE: coding agents for dexterous robotics	ICLR 2027 — under review
Co-first author / project co-lead [website]	

- Co-developed a robotics benchmark spanning **28 scenes and 5 embodiments**, with contact-rich manipulation, deformable objects, and task executions lasting up to **30 minutes**.
- Built a trajectory-generation pipeline with scene and strategy diversification, domain randomization, and replay verification; fine-tuned **SmolVLA across 8 simulation tasks** using **10–400 generated demonstrations per task**.
- Reconstructed the lab using **3D Gaussian splatting** and generated synthetic robot demonstrations in a simulation environment incorporating the reconstruction.
- Fine-tuned $\pi_{0.5}$ on the generated demonstrations and achieved **zero-shot sim-to-real deployment** on a Franka arm with a Robotiq gripper for grasping, unscrewing, and removing a light bulb.

Sampling-Guided Policy Search (SGPS)

ICRA 2027 — under review

First author [\[paper\]](#)

- Developed SGPS, combining recurring sampling-based MPC action refinement with first-order policy gradients to learn visual locomotion and manipulation policies on a **single RTX 4080**.
- Distilled specialists into one visual policy and transferred it **zero-shot to a real Go2** for autonomous trotting, crawling, hurdle traversal, and behavior switching using onboard depth, without online MPC.

Efficient visual reinforcement learning (SDPG)

CoRL 2026

Co-first author [\[paper\]](#) [\[code\]](#) [\[website\]](#)

- Co-led algorithm design and implemented the open-source codebase; estimates visual-policy gradients from perturbed rollouts without differentiating through rendering, using **about 1/5 of PPO's GPU memory**.
- Built visual RL benchmarks and led hardware experiments: trained a domain-randomized Go2 policy in **2 hours** and deployed it **zero-shot** on stairs, boxes, and uneven terrain.

Sampling-based hybrid mode control for agile quadruped skills

ICRA 2026

First author [\[paper\]](#) [\[website\]](#)

- Designed hybrid mode-sequence optimization with asymptotic-optimality guarantees; completed all three agile skills (foot stand, jump flip, handstand), with **~39% lower cost** than the best fixed-sequence baseline.
- Deployed **50 Hz** predictive control on a laptop-class CPU with EKF state estimation, achieving standing-to-handstand transitions on a real Go2 without motion capture.

Graduate Student Researcher, Carnegie Mellon University

Pittsburgh, PA

Advisor: Prof. Amir Barati Farimani

Aug 2020 – May 2022

- Developed an energy-saving snake-robot locomotion controller with an online model-based policy-gradient RL method in physics simulation; published in ASME J. Mechanisms and Robotics (2023).

TECHNICAL SKILLS

Programming & ML: Python, C++, PyTorch, JAX, Hugging Face Transformers; VLA fine-tuning ($\pi_{0.5}$, GR00T N1.6, SmolVLA), imitation learning (ACT, behavior cloning), visual RL, PPO.

Robotics & Control: ROS/ROS 2, MPC, motion planning, EKF, teleoperation, sim-to-real transfer; Franka, Robotiq, Unitree A1/Go2/G1.

Simulation & Infrastructure: MuJoCo/MJX, Isaac Sim/Lab/Gym, Genesis, Brax, 3D Gaussian splatting; LeRobot, Docker, Linux, Git, Weights & Biases, distributed training (torchrun).

SELECTED PUBLICATIONS & MANUSCRIPTS

Z. Shen*, H. You*, **Y. Liu***, et al. “[EmbodiedSWE: Coding Agents for Long-Horizon Dexterous Robotics](#)”. **ICLR 2027, under review** (*equal contribution and project leadership).

Y. Liu, H. You, Q. Wang, D. Rakita, I. Abraham. “[Accelerating Visual Policy Learning with Sampling-Based Model Predictive Control](#)”. **ICRA 2027, under review**.

H. You*, **Y. Liu***, D. Zong, et al. “[Efficient On-policy Visual-RL via Stochastic Decoupled Policy Gradient](#)”. CoRL 2026 (*equal contribution).

Y. Liu, H. You, I. Abraham. “[Sample-Based Hybrid Mode Control: Asymptotically Optimal Switching of Algorithmic and Non-Differentiable Control Modes](#)”. ICRA 2026.

H. You, **Y. Liu**, I. Abraham. “[Accelerating Visual-Policy Learning through Parallel Differentiable Simulation](#)”. NeurIPS 2025 (**Spotlight**).

C. Hughes, **Y. Liu**, et al. “[Asymptotically Optimal Ergodic Coverage on Generalized Motion Fields](#)”. RSS 2026.

Full list: [Google Scholar](#).

TEACHING & SERVICE

Teaching: Yale ENAS 773, Robotics, Control, and Learning (TF); CMU 18-461, Machine Learning (TA).

Reviewer: ICRA (2022–2026), IROS (2022–2026), RSS (2025), ECCV (2024), IEEE RA-L.